

Audio DSP & Communications Engineering Suite

Comprehensive Workflow & Signal Principles

Overview

This document serves as a comprehensive guide to the Digital Signal Processing (DSP) and Communications principles implemented in the Master Audio Lab.

1. Core Time-Domain Operations

Trimming: Extracts a specific time window from the audio array by slicing the discrete samples: `data[start_idx:end_idx]`.

Volume Scaling: A simple amplitude multiplication of the discrete signal by a scalar factor.

Time Reversal: Flips the discrete time sequence $x[n]$ to $x[-n]$. In Python, this is achieved via array slicing `data[::-1]`, which plays the audio backwards.

Echo: Modeled as a Finite Impulse Response (FIR) filter or convolution. The signal is delayed by a specific number of samples (`delay_ms * sr`) and added back to the original signal with an amplitude decay factor.

2. Frequency & Time Stretching

Resampling (Chipmunk Effect): Alters the sampling rate of the digital signal. When playback remains at the original sample rate, both the speed and pitch shift proportionally. We use Librosa for high-quality resampling.

Phase Vocoder: A sophisticated algorithm to stretch time without affecting pitch. It uses the Short-Time Fourier Transform (STFT) to analyze frequencies, scales the time axis, carefully unwraps and preserves the phase consistency between frames, and reconstructs the audio using the Inverse STFT.

Aliasing: Based on the Nyquist-Shannon Sampling Theorem. If a signal is downsampled without first applying a strict lowpass anti-aliasing filter, high frequencies wrap around (alias) into the lower frequency spectrum, causing irreversible metallic distortion.

3. Digital Filtering & Equalization

10-Band EQ: Implemented using Web Audio API Biquad Filters. These are Infinite Impulse Response (IIR) peaking filters that allow real-time boosting or cutting of specific frequency bands.

Digital Filters (Backend): Uses SciPys Butterworth filter design. Butterworth filters are chosen for their maximally flat frequency response in the passband, ensuring no ripple artifacts when applying Lowpass,

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Highpass, or Bandpass operations.

4. Communication Channel Simulation

Multipath / Inter-Symbol Interference (ISI): Simulates a physical environment where a signal reflects off buildings or mountains. This is implemented by convolving the audio with a channel impulse response (taps).

AWGN (Additive White Gaussian Noise): Simulates thermal noise in radio receivers. We calculate the signal power, generate Gaussian noise of corresponding power based on the desired Signal-to-Noise Ratio (SNR), and add it to the signal.

Single-Tone Jamming: Simulates an intentional or unintentional continuous wave (CW) interference. We generate a pure sine wave at a specific frequency (e.g., 1000 Hz) and add it to the audio at a specific SNR.

5. Receiver & Equalization

Zero-Forcing (ZF) Equalizer: The simplest equalizer. It takes the FFT of the signal and the channel taps, and divides the signal by the channel response ($W = 1/H$). While it perfectly removes multipath, it severely amplifies background noise in frequencies where the channel response is weak.

Minimum Mean Square Error (MMSE) Equalizer: A more robust approach that considers the noise floor. It balances inverting the channel and suppressing noise ($W = H^* / (|H|^2 + 1/SNR)$), yielding much better audio recovery in noisy environments.

6. Voice Biometrics (Speaker Matcher)

Feature Extraction: Uses Mel-Frequency Cepstral Coefficients (MFCCs). This mimics the human ears non-linear perception of pitch (the Mel scale) and extracts the spectral envelope (timbre) of the voice, ignoring pitch variations.

Identification: Computes the mean MFCC vector across time to create a Voice Print. To identify a speaker, it calculates the Cosine Similarity between the uploaded voice print and all stored voice prints in the SQLite database, returning the closest match.